

ADR 2: Model Hosting Strategy (Self-Hosted vs. API)

- **Status:** Accepted
- **Context:** The system processes live transaction descriptions containing sensitive counterparty details, account configurations, and high-value cross-border payment movements across India, the EU, and US markets. Regulatory bodies (such as the RBI and GDPR frameworks) heavily penalize third-party data leakage and require strict data localized residency.
- **Decision:** Utilize a **Self-Hosted Open-Source Model Strategy** powered by local inference utilities (Ollama/vLLM) utilizing **DeepSeek-R1-7B** for localized prototyping and **Qwen-2.5-72B-Instruct** for production hardware scale.
- **Alternatives Considered:** * *Proprietary Cloud APIs (OpenAI GPT-4o / Anthropic Claude 3.5 Sonnet):* Rejected. Although highly capable, transmitting transaction descriptions to external corporate APIs violates data residency laws and risks exposing proprietary banking metadata.
- **Consequences:** Shifts the operational burden from consumption-based variable pricing to fixed infrastructure expenses. Production environments will require provisioning enterprise-grade GPU clusters (e.g., NVIDIA H100/A100 instances) to host the 72B parameter reasoning engine within the regional Kubernetes environments.